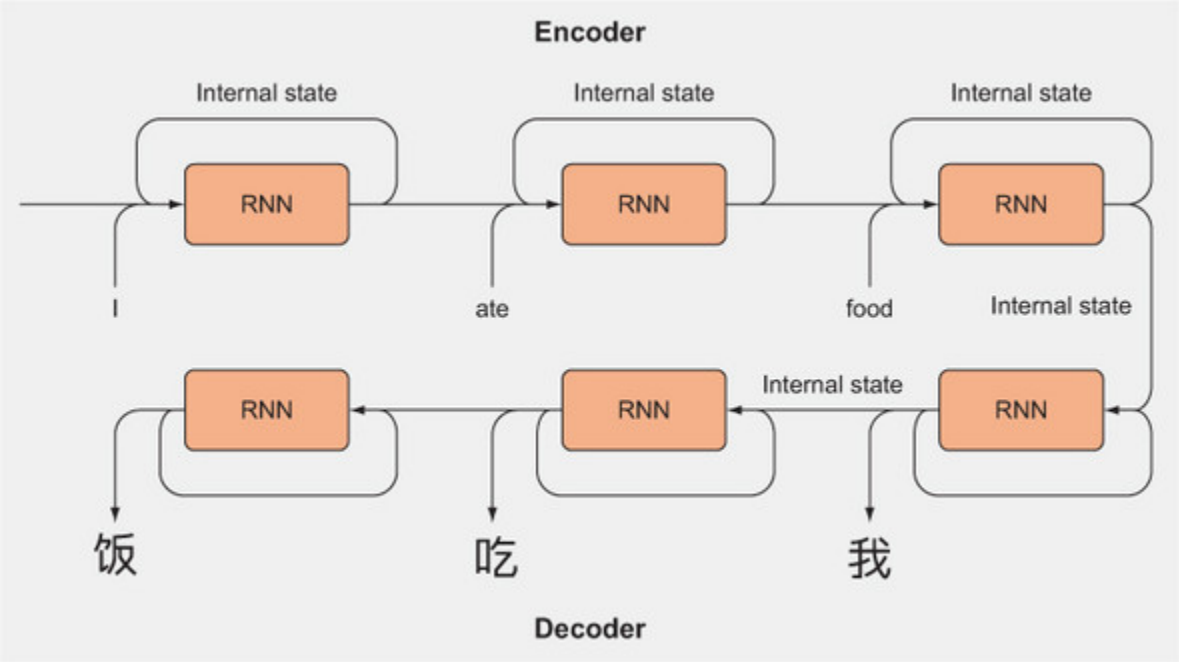
Figure 10.7. Schematic of an RNN language model. Two separate RNNs are used, an encoder and a decoder. The encoder takes an input sentence word by word and, once complete, sends its internal state to the decoder RNN, which produces each word in the target sentence until it halts.



While many thought that language models needed recurrence to work well, given the natural sequential nature of language, researchers found that a relatively simple attention model with no recurrence at all could perform even better and is trivially parallelizable, making it easier to train faster and with more data. These are the so-called *transformer models*, which rely on self-attention. We will not get into their details—we'll just sketch out the basic mechanism here.

The idea is that a Chinese word, c_i, can be translated as a function of the weighted combination of a context of English words, e_i. The context is simply a fixed-length collection of words that are in proximity to a given English word. Given the sentence "My dog Max chased a squirrel up the tree and barked at it," a context of three words for the word "squirrel" would be the subsentence "Max chased a squirrel up the tree" (i.e., we include the three words on either side of the target word).